

AMBUROUTE: SMART EMERGENCY DISASTER RESPONSE OS

Complete Engineering Report, Architectural Specifications, Technology Stack & Master Viva Handbook

1. Executive Summary & Problem Statement

English: In urban environments, ambulances face severe delays due to traffic gridlock, manual signal halts, and lack of pre-arrival ER coordination. The 'Golden Hour' (first 60 minutes) is critical in trauma and cardiac cases. **AmbuRoute** is an end-to-end intelligent IoT mission operating system featuring automated V2I (Vehicle-to-Infrastructure) traffic light preemption ('Green Wave'), real-time clinical triage, multi-criteria hospital matching, and bidirectional pre-arrival ER telemetry.

Hinglish Explanation: Real world me ambulance ko traffic jam aur red signals ki wajah se hospital pahunchne me bahut der ho jaati hai, jisse patient ki jaan ja sakti hai. Is project (AmbuRoute) ka main goal hai ambulance ke raste ke saare signals ko IoT se automatically GREEN karna, patient ke vitals hospital ER ko advance me bhej kar bed reserve karwana, aur sabse best hospital select karna.

2. Complete Technology Stack (A to Z Breakdown)

Layer / Module	Technology Used	Role & Practical Justification
Frontend UI	React 19, Vite	High-performance Single Page Application with sub-second hot reload and zero rendering
Styling System	Tailwind CSS + Vanilla CSS	Ultra-clean Red & White Emergency theme + Tactical Dark Mission mode with responsive
Mapping & GIS	Leaflet.js + OpenStreetMap / Esri	Zero-cost open mapping with custom DivIcon markers, animated radar pulses, and corridor
Backend Server	FastAPI (Python 3.11), Uvicorn	High-throughput asynchronous ASGI microservice with native Pydantic data validation.
Real-Time Duplex	WebSockets (FastAPI Pub/Sub)	Continuous 10 Hz bidirectional telemetry push between ambulance MDT, hospital desk, and
Audio Engine	Web Audio API (Synthesizer)	Pure browser-native sound generation for sirens, clicks, and green-wave pings (zero audio
Routing Engine	Haversine Geodesic Math + Graph	Accurate distance calculation, waypoint interpolation, and dynamic intersection proximity d
Decision Matrix	TOPSIS / Multi-Criteria Scoring	Weighted AI ranking: Distance (30%) + Bed Vacancy (30%) + Cath Lab (25%) + ER Wait (10%)
Triage AI	Rule-Based Clinical Risk Model	Real-time hemodynamic instability scoring, Shock Index, and Killip Class categorization.

3. System Architecture & Operational Workstations

English: The system employs a modern **Role-Based Access Control (RBAC)** architecture with 4 distinct operational portals:

Role Portal	Default Credentials	Operational Workstation & Capabilities
■ Ambulance Driver / Paramedic	driver108 / 1080	Live GPS turn-by-turn navigation, dynamic hospital selector, real-time vitals entry, and siren c
■ Hospital ER Doctor / Staff	doctor_aiims / aiims123	Pre-arrival trauma bay reception, real-time Lead-II ECG waveform monitor, ICU bed reservati
■ Traffic Police Controller	traffic_gkp / traffic123	Physical 3-aspect traffic light console (Red/Amber/Green), Master Green Wave override, and
■ Super Administrator (God Mode)	admin / admin123	Full root authority: Speed slider (0-120 km/h), patient vitals injector, signal override, hospital b

Hinglish Explanation: Examiner agar pooche ki 4 log kaise kaam karte hain: Ambulance Driver raste aur patient par dhyan deta hai, Hospital Doctor aane wale mareez ke liye ICU bed arrange karta hai, Traffic Police signals control karti hai, aur Administrator project me koi bhi value live change kar sakta hai.

4. Step-by-Step Project Working (Data Flow)

- **Step 1 (Incident Dispatch):** 108 CAD operator inputs or receives emergency call (e.g. STEMI, Poly-Trauma, Pediatric Sepsis). Ambulance MEDIC-12 is mobilized.
- **Step 2 (Hospital Recommendation):** System runs TOPSIS AI matrix comparing AIIMS, BRD Medical, and Fatima Hospital based on ICU beds, travel time, and cath-lab capability.
- **Step 3 (V2I Traffic Preemption):** As ambulance drives, GPS telemetry triggers optical/radio preemption handshakes 500m ahead, changing intersections to Green Wave.
- **Step 4 (ER Telemetry & Bed Reservation):** Paramedics enter live vitals (HR, BP, SpO2) which stream to Hospital ER Trauma Desk via WebSockets, allowing doctors to reserve ICU beds before arrival.

5. Master Viva Voce Q&A; Handbook (English + Hinglish)

Prepare these high-frequency questions to excel in your final year project presentation:

Q1: What is the main innovation of AmbuRoute compared to Google Maps?

Technical Answer (EN): Google Maps provides static navigation for general public but cannot communicate with traffic lights or transmit medical telemetry to hospitals. AmbuRoute actively preempts traffic signals (Green Wave) via V2I protocols and transmits pre-arrival clinical vitals to the ER bay.

Aasan Bhasha Me (Hinglish): Google Maps sirf rasta batata hai, par AmbuRoute traffic signals ko automatically GREEN karta hai aur hospital me doctor ko mareez ke live vitals bhejta hai taaki bed pehle se ready rahe.

Q2: How does the V2I Traffic Signal Preemption work technically?

Technical Answer (EN): The ambulance broadcasts its GPS coordinates and heading at 10 Hz over 5G C-V2X / DSRC. When within the 500m geofence buffer of an intersection, the controller calculates ETA and initiates cross-traffic yellow clearance, holding green priority until the ambulance clears.

Aasan Bhasha Me (Hinglish): Ambulance jab kisi signal se 500 meter door hoti hai, to system C-V2X IoT handshake bhejta hai. Signal cross-traffic ko rokk kar ambulance ke liye green wave chalu kar deta hai.

Q3: How does the Hospital Matrix decide which hospital to select?

Technical Answer (EN): It utilizes a Multi-Criteria Decision Making (MCDM) weighted TOPSIS algorithm: Travel Distance (30%), Bed Availability (30%), Cath Lab / Trauma Level (25%), and ER Wait Efficiency (15%).

Aasan Bhasha Me (Hinglish): Ye algorithm sirf doori nahi dekhta; ye dekhta hai ki kis hospital me ICU bed khali hai aur specific specialist doctor (jaise Cath Lab) available hai ya nahi.

Q4: Why did you use WebSockets instead of normal HTTP REST APIs?

Technical Answer (EN): REST APIs require polling (request/response overhead) with 1-2 second latency. WebSockets maintain a persistent, full-duplex TCP socket, enabling real-time telemetry streaming at under 15ms latency.

Aasan Bhasha Me (Hinglish): HTTP me har baar request bhejni padti hai jisse delay hota hai. WebSocket se continuous direct connection rehta hai, jisse ambulance ki speed aur ECG bina rukavat hospital screen par dikhti hai.

Q5: What is the role of the Super Administrator deck?

Technical Answer (EN): It provides God-Mode authority for demonstration and administrative testing: live speed adjustment, patient vital injection, full corridor green wave override, hospital bed modification, and traffic jam simulation.

Aasan Bhasha Me (Hinglish): Administrator panel se project ki kisi bhi value (speed, BP, heart rate, signals, beds) ko examiner ke samne live change karke dikhaya ja sakta hai.

Q6: How is patient triage calculated in the AI Clinical module?

Technical Answer (EN): The model computes the Shock Index (HR / Systolic BP), GCS neurological coma score, and oxygen desaturation to stratify patients into Immediate (Red), Urgent (Yellow), or Delayed (Green).

Aasan Bhasha Me (Hinglish): System Heart Rate aur Blood Pressure ka ratio nikaalta hai (Shock Index) aur GCS score check karke patient ko critical red ya yellow tag deta hai.

Q7: How are audio sirens generated without audio files?

Technical Answer (EN): Through the HTML5 Web Audio API, which synthesizes sound waves (sine, square, and triangle oscillators) directly on the client CPU, resulting in zero network bandwidth and zero audio file dependencies.

Aasan Bhasha Me (Hinglish): Web Audio API browser ke andar mathematical frequency banakar siren aur clicks bajati hai, jisse koi mp3 file download karne ki zaroorat nahi padti.

Q8: What happens if a traffic signal fails to connect to IoT?

Technical Answer (EN): The system defaults to fail-safe mode: it alerts the ambulance of the connection failure, highlights the intersection in red on the GIS map, and allows the traffic police officer to execute a manual override.

Aasan Bhasha Me (Hinglish): *Agar IoT signal disconnect ho jaye, to system map par alert dikhata hai aur Traffic Police desk par manual switch button enable kar deta hai.*

6. Git Push & 100% Free Deployment Guide

Follow these exact steps to push code to GitHub and host live for free:

Step A: Push Project to Your GitHub

```
git init
git add .
git commit -m "feat: Complete AmbuRoute Emergency OS with Role Portals & Admin Deck"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/amburoute.git
git push -u origin main
```

Step B: Deploy Frontend (Vercel - 100% Free)

1. Open **vercel.com** and sign in with GitHub.
2. Click **Add New Project** and import your **amburoute** repository.
3. Set Root Directory to **frontend**.
4. Click **Deploy**. Your frontend will be live in 60 seconds with a free `.vercel.app` HTTPS domain!

Step C: Deploy Backend (Render.com - 100% Free)

1. Open **render.com** and create a free account.
2. Click **New +** -> **Web Service** -> Connect GitHub repository.
3. Set Root Directory to **backend**.
4. Build Command: `pip install -r requirements.txt`
5. Start Command: `uvicorn app.main:app --host 0.0.0.0 --port $PORT`
6. Click **Create Web Service**. Backend API and WebSockets are live!